

BrokerVerse OOTB

Insurance broking platform by iorta TechNXT

Solution Architecture · Document 04 of 11

Technology Stack

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Version	Date	Author	Change
1.0	29 September 2026	iorta TechNXT	Initial issue
1.1	29 September 2026	iorta TechNXT	Updated to the current system: placement journey, co-insurance, posting rules, period end and BIR tax, bank reconciliation, document numbering, commission rate matrix, single PDF engine, uploads and go-live imports, housekeeping, business time zone, broker roles, deployment on the BrokerVerse URL

Review and approval

Role	Name	Signature	Date
Solution architect			
DevOps / infrastructure lead			
Information security			
Business owner			

Conventions

- Statements of fact are derived from the code, configuration and database of the baseline above (file paths are given relative to the repository). Figures measured on the local test system are labelled **test data** or **indicative**.
- Targets, sizes, retention periods and thresholds that are not facts of the code are labelled **Recommended / to be confirmed by the business and DevOps**.
- Items marked **Gap** are defects or missing capabilities found while preparing this document.

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1 Introduction

1.1 Purpose and scope

This document lists the technologies used by BrokerVerse OOTB, with the versions resolved in the lock files of the baseline, their purpose and their licence type, together with the runtime, infrastructure, build and test tooling. It is the reference for dependency management, security patching and licence review.

Sources: `backend/package.json` and `backend/package-lock.json`, `brokerverse/package.json` and `brokerverse/package-lock.json` (licences read from the installed packages), `backend/Dockerfile`, `brokerverse/Dockerfile`, `brokerverse/nginx.conf`, `docker-compose.yml`, `.github/workflows/ci.yml`, `brokerverse/.github/workflows/deploy.yml` and `deploy/README.md`. Versions are the exact versions in the lock files; the manifests use caret ranges and `npm ci` installs the locked versions.

1.2 Stack overview

Layer	Technology
Browser	Current evergreen browsers (build targets >0.2%, not dead, not op_mini all)
Front end	React 18.2 single-page application, PrimeReact 10, PrimeFlex, PrimeIcons, Redux Toolkit 2, React Router 6, Formik, i18next, axios, Chart.js, FullCalendar, Sass; built with Create React App 5 (react-scripts) and craco 7
Web serving	Amazon S3 and CloudFront on the existing BrokerVerse URL; nginx 1.27-alpine in the Docker Compose option
API	Node.js 22 (ES modules), Express 4.22, node-postgres 8, zod 3, pino 9, helmet 8, jsonwebtoken 9, bcryptjs 2, multer 1.4, nodemailer 10, node-cron 4
Database	PostgreSQL 16 (16.13 on the build host; <code>postgres:16-alpine</code> in Docker Compose and CI; Amazon RDS for PostgreSQL 16 as the production option) with <code>pgcrypto</code> ; database time zone Asia/Manila
File storage	Persistent volume at <code>UPLOAD_DIR</code> (Amazon EFS or another shared volume in production)
E-mail	Office 365 SMTP (<code>smtp.office365.com</code> , port 587, STARTTLS) through <code>SMTP_URL</code>
Containers	Docker images <code>node:22-alpine</code> (API) and a <code>node:22-alpine</code> build stage with an <code>nginx:1.27-alpine</code> runtime (web); Docker Compose
CI/CD	GitHub Actions: <code>.github/workflows/ci.yml</code> (backend lint and tests on PostgreSQL 16, front-end tests and build, backend image build) and <code>brokerverse/.github/workflows/deploy.yml</code> (front-end tests, build, S3 sync, CloudFront invalidation)
Cloud	AWS ap-southeast-1 (Singapore): S3 and CloudFront for the front end; for the backend a load balancer with HTTPS, a container service (ECS Fargate, App Runner or EC2), PostgreSQL, a persistent volume and a secret store (Secrets Manager or SSM); monitoring recommended in CloudWatch

2 Backend (backend/)

2.1 Runtime dependencies

Package	Version	Licence	Purpose in BrokerVerse
express	4.22.3	MIT	HTTP server, routing, middleware chain (<code>src/app.js</code> , module routers)
pg (node-postgres)	8.23.0	MIT	PostgreSQL client and connection pool (<code>src/db/pool.js</code> , 10 connections), type parsers for numeric, bigint and date
zod	3.25.76	MIT	Request validation schemas (<code>lib/validate.js</code>)
pino	9.14.0	MIT	Structured JSON logging with redaction (<code>lib/logger.js</code>)
pino-http	10.5.0	MIT	HTTP access log with request id
helmet	8.3.0	MIT	Security headers
cors	2.8.6	MIT	Cross-origin policy from <code>CORS_ORIGINS</code>
jsonwebtoken	9.0.3	MIT	Access, refresh, challenge and approval tokens (HS256)
bcryptjs	2.4.3	MIT	Password hashing (bcrypt in JavaScript, cost 10)
multer	1.4.5-lts.2	MIT	Multipart uploads held in memory with size and count limits
nodemailer	10.0.12	MIT-0	SMTP delivery of the e-mail outbox
node-cron	4.6.0	ISC	Cron scheduling of the jobs, in the business time zone
dotenv	16.6.1	BSD-2-Clause	Loads <code>.env</code> in development

The backend has 13 direct runtime dependencies (333 packages in the lock file, including development tools). The following are part of the code base instead of third-party libraries: the PDF engine (`lib/pdf`: writer, layout, tables, fonts, images) with the letterhead (`lib/letterhead.js`), the XLSX writer (`lib/xlsx.js`) and ZIP container (`lib/zip.js` over `node:zlib`), CSV with formula-injection guarding (`lib/csv.js`), TOTP two-factor (`lib/totp.js`), AES-256-GCM secret encryption and HMAC link signing (`lib/secrets.js` over `node:crypto`), rate limiting (`lib/rateLimit.js`) and workbook import with inflate caps (`lib/uploadLimits.js`).

2.2 Development and test dependencies

Package	Version	Licence	Purpose
vitest	2.1.9	MIT	Integration tests against a real PostgreSQL (<code>backend/test</code> , 50 files, 518 tests)
supertest	7.3.0	MIT	HTTP assertions against the in-process app
eslint, @eslint/js	9.39.5	MIT	Linting (<code>npm run lint</code> ; <code>no-console</code> outside scripts, <code>no-unused-vars</code> and <code>eql</code> as errors)
globals	15.15.0	MIT	ESLint environment globals

2.3 Scripts and tools

Command	Purpose
<code>npm start</code>	<code>node src/server.js</code> : production configuration check, migrations under an advisory lock, seed, HTTP server, scheduler
<code>npm run dev</code>	Development server with <code>node --watch</code>

Command	Purpose
<code>npm run migrate, npm run seed, npm run db:reset</code>	Apply migrations; run the seed; drop and rebuild the schema (development only)
<code>npm run export:api</code>	Write OpenAPI, Postman and Excel API documentation from the route registry
<code>npm run purge:sample</code>	Remove sample data before go-live (dry run by default)
<code>npm run check:settings</code>	Compare the setting keys read in code with the keys in the database
<code>node scripts/provision-users.js</code>	Create named users from a CSV kept outside the repository
<code>node scripts/build-upload-templates.js</code>	Regenerate the upload templates in <code>docs/templates</code>
<code>node scripts/fk-index-report.js</code>	List foreign keys without a supporting index
<code>npm test, npm run lint</code>	Tests and lint

3 Front end (brokerverse/)

3.1 Runtime dependencies

Package	Version	Licence	Purpose
react, react-dom	18.2.0	MIT	UI library
react-scripts (Create React App)	5.0.1	MIT	Build tool chain (webpack, Babel, Jest, ESLint)
@craco/craco	7.1.0	Apache-2.0	Overrides the CRA webpack configuration (<code>craco.config.js</code>)
primereact	10.3.1	MIT	Component library (tables, forms, dialogs, toasts, calendars)
primeflex	3.3.1	MIT	CSS utility classes
primeicons	6.0.1	MIT	Icon font
@reduxjs/toolkit	2.0.1	MIT	Application state (<code>src/redux/store.js</code>)
react-redux	9.0.4	MIT	React bindings for Redux
react-router, react-router-dom	6.30.6	MIT	Client-side routing (<code>routes/MainRoute.js</code>)
history	5.3.0	MIT	Navigation history helper
formik	2.4.5	Apache-2.0	Forms and validation
axios	1.20.0	MIT	HTTP client (interceptor with token refresh)
i18next, react-i18next, i18next-browser-languagedetector	25.8.13 / 16.5.4 / 8.2.1	MIT	Internationalisation (English, Thai)
chart.js	4.5.1	MIT	Dashboard charts
@fullcalendar/core, daygrid, react	6.1.x	MIT	Calendar views (open items)
moment	2.29.4	MIT	Date handling in four older screens
sass	1.69.5	MIT	SCSS styles and theme
@fontsource/nunito	5.3.0	OFL-1.1	Nunito web font, bundled (no external font service)
redux-logger	3.0.6	MIT	Redux logging in development builds only
js-cookie, react-pro-sidebar, web-vitals	3.0.8 / 1.1.0-alpha.1 / 2.1.4	MIT, MIT, Apache-2.0	Listed in <code>package.json</code> but no longer imported; to be removed with the next dependency update

The `@testing-library/*` packages are listed under dependencies but are used only by the 30 front-end tests.

3.2 Build and delivery

Item	Detail
Build	<code>npm run build</code> (<code>craco build</code>). <code>REACT_APP_BASE_URL</code> (the API base, for example <code>https://<brokerverse-url>/api</code>) is compiled in at build time; <code>GENERATE_SOURCEMAP=false</code> .

Item	Detail
Deployment workflow	<code>brokerverse/.github/workflows/deploy.yml</code> : on a push or pull request to <code>dev</code> , Node 22, <code>npm ci --legacy-peer-deps</code> , <code>craco test</code> , a check that the repository variable <code>REACT_APP_BASE_URL</code> is set (push only), build; on a push to <code>dev</code> the deploy job syncs <code>build/</code> to the S3 bucket with <code>--delete</code> and invalidates CloudFront <code>/*</code> . Secrets: <code>AWS_ACCESS_KEY_ID</code> , <code>AWS_SECRET_ACCESS_KEY</code> , <code>S3_BUCKET</code> , <code>CLOUDFRONT_DISTRIBUTION_ID</code> .
Repository CI	<code>.github/workflows/ci.yml</code> : on every push to <code>brokerverse-platform</code> or <code>main</code> and on pull requests; backend <code>npm ci</code> , <code>eslint</code> , <code>vitest</code> against a <code>postgres:16-alpine</code> service; front end <code>npm ci</code> , <code>craco test</code> , production build; backend image build after the backend job passes.
Container	Multi-stage <code>brokerverse/Dockerfile</code> : <code>node:22-alpine</code> build, <code>nginx:1.27-alpine</code> runtime with <code>nginx.conf</code> (SPA fallback, <code>/static/</code> cached for one year, <code>index.html</code> not cached, <code>/api/</code> proxied to the API, <code>client_max_body_size 25m</code> , security headers).

4 Infrastructure and platform

Component	Version / service	Purpose and status
Node.js	22 (image node:22-alpine)	API runtime and both GitHub workflows; engines in backend/package.json is >=22
PostgreSQL	16	System of record; time zone Asia/Manila (documents 07 and 09)
nginx	1.27-alpine	Web container of the Docker Compose option
Docker, Docker Compose	Compose file without a version key	Single-server, test and trial installations
GitHub Actions	actions/checkout@v4, setup-node@v4, upload-artifact@v4, download-artifact@v4, configure-aws-credentials@v4	CI and front-end deployment
AWS	ap-southeast-1	S3 and CloudFront for the front end; backend services per deploy/README.md (load balancer, container service, PostgreSQL, persistent volume, secret store)
Office 365	SMTP submission	Outbound e-mail from connect@iortatechnxt.com
Monitoring	CloudWatch (recommended)	Logs, metrics, alarms (document 11)

5 Lifecycle and risk notes

Item	Observation (from docs/review/CODE_REVIEW.md and the lock files)
Backend advisories	<code>npm audit</code> reported no backend advisories after the nodemailer 10 and node-cron 4 upgrades (29 Sep 2026).
Front-end advisories	71 advisories were reported at the review (3 critical, 35 high), almost all build-time dependencies of react-scripts 5.0.1; react-router 6 and craco have moderate ones. Create React App is no longer maintained. A move to Vite (and later React Router 7) is the planned remedy; to be scheduled.
Pipeline credentials	The deployment workflow uses long-lived AWS access keys stored as GitHub secrets. Recommended: GitHub OIDC with an IAM role limited to the bucket and the distribution.
Backend release pipeline	CI builds the backend image but does not push it to a registry. The image registry and the release steps are to be confirmed by DevOps.
Unused front-end packages	<code>js-cookie</code> , <code>react-pro-sidebar</code> and <code>web-vitals</code> are no longer imported; <code>moment</code> is used by four files. Remove them with the next dependency update.
Licences	All listed runtime dependencies use permissive licences (MIT, MIT-0, ISC, BSD-2-Clause, Apache-2.0) and the Nunito font the SIL Open Font License; no copyleft licence was found among the direct dependencies. A full scan of indirect dependencies (for example license-checker) is recommended before each release.

6 Related documents

This document is one of eleven that together form the BrokerVerse OOTB Solution Architecture. They are maintained and issued together (source and build: [docs/architecture/README.md](#)).

No.	Document	File
01	Solution Component Diagram and Application Architecture	01_BrokerVerse_Solution_Component_Diagram_Application_Architecture.pdf
02	Database Design	02_BrokerVerse_Database_Design.pdf
03	Database Inventory	03_BrokerVerse_Database_Inventory.pdf
04	Technology Stack (this document)	04_BrokerVerse_Technology_Stack.pdf
05	Shared Service Components	05_BrokerVerse_Shared_Service_Components.pdf
06	Capacity and Performance	06_BrokerVerse_Capacity_and_Performance.pdf
07	High Availability and Resiliency	07_BrokerVerse_High_Availability_and_Resiliency.pdf
08	RTO and RPO	08_BrokerVerse_RTO_and_RPO.pdf
09	Backup and Recovery	09_BrokerVerse_Backup_and_Recovery.pdf
10	Data Archival, Housekeeping and Restoration	10_BrokerVerse_Data_Archival_Housekeeping_and_Restoration.pdf
11	Monitoring	11_BrokerVerse_Monitoring.pdf

7 Glossary

Terms and abbreviations used in this document.

Term	Meaning
Advisory lock	PostgreSQL application-level lock (<code>pg_try_advisory_lock</code> , <code>pg_advisory_lock</code>) used so a job runs on one API instance at a time and only one instance applies migrations.
API	Application programming interface; here the BrokerVerse REST API under <code>/api</code> served by the Node.js backend.
bcrypt	Adaptive password-hashing function (bcryptjs library) used for user passwords.
CloudFront	AWS content delivery network that serves the single-page application and can route <code>/api</code> to the backend.
CloudWatch	AWS monitoring service: logs, metrics, dashboards and alarms.
CORS	Cross-Origin Resource Sharing: browser rule set that the API restricts to the configured web origins (<code>CORS_ORIGINS</code>).
CRA / craco	Create React App (react-scripts) build tool chain, customised with craco (Create React App Configuration Override).
Cron	Time-based schedule expression (minute hour day month weekday) used by the scheduled jobs.
ECS / Fargate	Elastic Container Service (AWS) and its serverless compute mode for running the API containers.
EFS	Elastic File System (AWS): shared, multi-AZ network file system proposed for the upload directory.
HMAC	Keyed-hash message authentication code; signs file links and hashes one-time codes.
Housekeeping	Daily job that deletes operational rows (job runs, sent e-mails, sign-in history, expired tokens, read notifications) after the retention days in System Settings.
HS256	HMAC-SHA256 JSON Web Token signature algorithm, the only one the API accepts.
nginx	Web server used in the web container to serve the SPA and forward <code>/api</code> to the API container.
Office 365	Microsoft 365 mail service; BrokerVerse sends e-mail through <code>smtp.office365.com</code> with the mailbox in <code>SMTP_URL</code> .
RDS	Amazon Relational Database Service, the managed PostgreSQL target.
RPO	Recovery point objective: the maximum acceptable data loss, measured in time.
RTO	Recovery time objective: the maximum acceptable time to restore service.
S3	Amazon Simple Storage Service: object storage hosting the SPA build (and proposed archive store).
Seed	Idempotent data load run on every API start: reference data always, sample data only when <code>SEED_SAMPLE_DATA</code> is on.
SMTP	Simple Mail Transfer Protocol; the API sends e-mail through the server in <code>SMTP_URL</code> .
SPA	Single-page application: the React front end served as static files.
TOTP / 2FA	Time-based one-time password (RFC 6238) used for two-factor authentication.
zod	TypeScript-first schema validation library used to validate request bodies and parameters.